Adventist University of Central Africa
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COURSE SYLLABUS

I. General Information		
1. Course Information		
a.	Faculty:	Information Technology
b.	Department:	All
c.	Course Code:	MATH8213
d.	Course Name:	Multivariable Calculus and Differential Equations
e.	Number of credits:	4
f.	Prerequisite:	Applied Mathematics
g.	Academic Year	2025-2026
h.	Semester:	One
i.	Time:	Friday:8:00-16:00 , Monday: 8:00-12:00 Sunday :11:00-15:00
j.	Venue:	Gishushu Campus
k.	Technical Support:	
2. Instructors Information		
a.	Name:	TheonesteHakizimana
b.	Contact :	htheonest2000@yahoo.fr; Tel:0788592204
c.	Consultation Time:	Monday:12.30-14:00 Sunday:8:00-10:00
d.	Location:	Room:203

II. Course Description

This course intends to equip students with critical thinking and problem solving skills. It introduces students to basic concepts and various techniques used in Multivariable calculus and Differential Equations. It provides also mathematical tools used to describe and to find analytical solutions to world real problems connected with multivariable calculus and ordinary differential equations.

III. Learning Outcomes

a	Use 2D&3D Analytic Geometry techniques to solve Multivariable Calculus problems
b	Use Multivariable Calculus in solving optimization problems and evaluating lengths, surface areas and volumes of 2D and 3D Shapes
c	Use Differential equations methods in solving problems related to information and telecommunication
d	Equip students skills of critical thinking and solving problems related to information technology and Programming using mathematical approaches

IV. Indicative Content

Chapter 1: Analytic Geometry in 2D and 3D…

Week	Date	Objectives	Content	Learning Activities
1		Solve elementary Problems of 2D analytic Geometry	Elementary problems Planimetryy	Calculate lengths, angles and area of 2D shapes
		Solve geometrical problems of Line and Circle	Geometry of Line and Circle	Write down various equations of line and circle
2		Solve problems related to Conics	Second Order Curves	Write down various equations of line and circle
3		Solve elementary Problems of 3D analytic Geometry	Elementary problems of Stereometry	Calculate lengths, angles, area and volume of 3D polyedra
		Solve geometrical problems of the Line and plane	Geometry of Line, plane in 3D	Write down various equations of line and plane
4		Solve problems related to quadric surfaces	Second Order Surfaces	Write down canonical equations of quadric surfaces and construct them using local materials

Chapter 2: Multivariable Calculus

Week	Date	Objectives	Content	Learning Activities
5		Define and Characterize a Multivariable function	Overview on Multivariable function	Determine the domain range and Graph of a multivariable function
6		Find partial derivatives of a multivariable function	Partial Differentiation	Perform partial differentiation of a2 and3 variables function
7		Solve problems of optimization via Partial differentiation	Applications of Differentiation	Transform an optimization word problem into mathematical model and solve it
8		Evaluate double integrals	Integration: Double integrals	Evaluate double integrals on various regions of integration
9		Evaluate triple integrals	Integration: Triple integrals	Evaluate triple integrals on various regions of integration
			Applications of integration	

Chapter 3: Differential EquationS

Week	Date	Objectives	Content	Learning activities
10		Define and Characterize a Differential Equation	Over view on Differential Equations	Classify O Des in terms of Order and degree

		Solve ODE using algebraic meth	First Order ODEs	Solve separable, homogeneous, Exact, Linear and Bernoulli Equations
11		Solve ODE using analytic and algebraic methods	Second Order ODEs	Solve Linear DE with constant coefficient
12		Apply differential equations in solving word problems	Applications of ODES	Solve word problems via ODEs

V. Methodological Strategies

1. Cognitive	
a.	Various worked problems/exercises will give students the chance to develop their theoretical understanding and problem solving skills.
b.	Efforts will be made for active participation of students and group discussions will be promoted.
c.	5E model made by Engage/Excite, Explore, explain , Elaborate and Evaluate phases will be implemented
2. Application	
a.	Many problems taken from the real and daily life will be treated.
b.	Students will be shown the role of Mathematics in STEM education and ICT.
c.	Practical work related to make 2d and 3D shapes using local materials and Various ICT software (GeoGebra, Math graph, Math Calculator, Maple ,..)will be used during learning and teaching activities
3. Integration of faith and learning	
a.	Various examples showing that Mathematics does not contradict faith will be provided during Learning and teaching process
b.	Mathematics ethics promotes accuracy, non-contradiction and Exclusion of the Middle
c.	Biblical interpretation of some numbers:1,3,7,12,144,etc

VI. Students Requirements

1. Introduction of Class and University Policies

- Actively participation is encouraged
- Deadlines should be respected.
- Avoid cheating and plagiarism

2. Class Management

a.	Attendance:	Daily attendance and respect of punctuality
b.	Discipline:	Reinforce the discipline
c.	Code of Conduct:	
d.	Financial Clearance:	

VII. Inclusive Education

Ensure that All student can access teaching and learning materials, are participating in lessons and University life, and they are achieving learning outcomes and are integrating into the AUCA University Community. Students should be given opportunities to fully participate and achieve. Promote equal presence, equal participation and equal achievement

VIII. Assessment Strategies

Assignment, Quizzes, Participation, Projects :	30%
Mid Semester:	30%
Final Exam:	40%
Total	100%

IX. Strategy for Feedback and Student Support

Various worked problems/exercises will give students the chance to develop their theoretical understanding and problem solving skills. Efforts will be made for active participation of students. Quizzes followed by feedbacks will be provided. Students will be encouraged to visit library and integrate ICT in their learning and teaching process.

Consultations hours will be given to students

X. References

1. Text Books

(Write a text here)

Core Texts (These are available at the reference section of AUCA library)

-Stewart J. (2003), *Calculus, Concepts and contexts.3*, Metric International Version, Mc Master University

-Dineen S., (2000), *Function of two variables*, 2nd Edition, Chapman and Hall.

Background Texts

1.**Clapham Ch., Nicholson** J, (2009) *The concise dictionary of mathematics*, Oxford Press, 4th Edition, **ISBN 9780199235940**

2.**Smith R.T., Roland B.M**., (2002) *Multivariable Calculus*, second Edition, Mc Graw

3.**Riley K.F**., (2002) *Mathematical Methods for physics and engineering, a comprehensive guide*,

4.Rosen H.B. (2005), *Applied Calculus*, 8th Edition, Mc Graw-Hill

5. **Snehanshu Saha**, Ordinary Differential Equations: A Structured Approach, India, 2011.

6.**Hartmut Logemann**, Eugene P. Ryan, Ordinary Differential Equations, Analysis, Qualitative

2. Journals

3. Online Resources

http://tutorial.math.lamar.edu/Classes/DE/Linear.aspx,
http://tutorial.math.lamar.edu/Classes/DE/IntroFirstOrder.aspx :Linear first order ODE
http://tutorial.math.lamar.edu/Classes/DE/IntroSecondOrder.aspx: Linear second order ordinary differential equations

http://www.sosmath.com/diffeq/diffeq.htmlwww.mit.edu